



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: B.E.VI, Computer Science and Engineering

Course Code & Title: CCS339 Cryptocurrency and Blockchain technologies

Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit IV / Solidity

Course Outcome: CO 4

Unit Learning Outcome: UO 4a

Activity Chosen: Demonstration and Practice

Justification:

Hands-on training, using the Remix IDE for educational purposes, is a highly effective method for teaching specific tasks by providing real-world experience. Remix IDE, a popular tool for developing and testing smart contracts on the Ethereum blockchain, allows students to directly interact with the material they are learning. By engaging with the Remix IDE, students can write, deploy, and debug smart contracts, gaining a tangible understanding of blockchain technology and its applications. This method not only offers clear, practical examples of how smart contracts are developed and executed but also forces students to reconsider their prior understanding of blockchain concepts.

Time Allotted for the Activity: 20 minutes

Details of the Implementation:

RemixIDE

- Students were instructed to open a web browser and navigate to the official Remix IDE website.
- They were introduced to the default "Solidity" environment for smart contract development.
- Students created a new file and named it with the extension.sol.
- They explored various data types in the Solidity programming language and practiced by creating "Hello World" and addition programs.
- Additionally, students learned how to write smart contracts and understand the deployment process

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PSO2	PSO3
CO4	2	2	2	2	3	2	2

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO5	PSO2	PSO3
	2	2	2	2	3	2	2
Justification for correlation	Students understood the concept of blockchain technology with Engineering Knowledge	Students analyzed blockchain use cases and scenarios using engineering sciences.	Students developed blockchain-based solutions by implementing smart contracts.	Students investigated complex problems related to blockchain models and explore potential improvement	Students effectively used blockchain development tools and platforms to implement and gain hands-on experience in blockchain technology.	Students will be able to apply blockchain concepts to develop reliable IT solutions.	Students will be able to incorporate blockchain with AI to provide solutions to real-world problems in Industry.

Screenshot of the practice:



Fig:1 Demonstration of RemixIDE



Fig:2 Demonstration of solidity Compiler

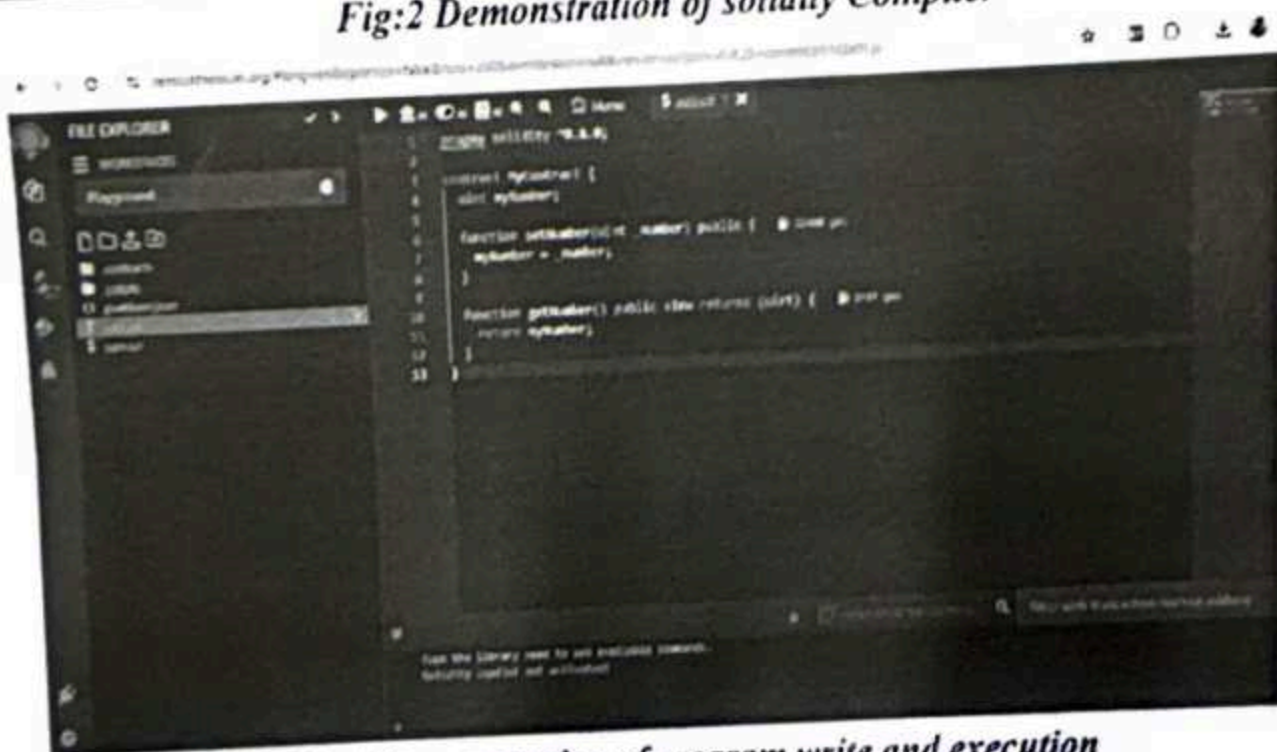


Fig:3 Demonstration of program write and execution

Reflective Critique:

- Students know the possible online tools related to the subjects.
- The demonstration activity enhances students' awareness of online tools relevant to the subject matter, expanding their resource base for future learning endeavors.
- Introducing students to a variety of online tools cultivates adaptability and prepares them for diverse technological environments.

Benefit of the practice:

- The students enjoyed the activity.
- Hands-on practice aids in depth about concept clearly.
- Active engagement promotes experiential learning, enabling students to adopt theoretical knowledge and apply it to practical scenarios with greater confidence.

Challenges faced in implementation:

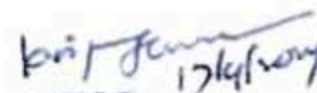
- Remix IDE is designed as an online environment, and its integration with local development environments is limited. So, not able to deploy in local environment.

References:

1. <https://andersbrownworth.com/blockchain/>
2. <https://remix.ethereum.org/#lang=en&optimize=false&runs=200&evmVersion=nu11&version=soljson-v0.8.22+commit.4fc1097e.js>
3. <https://www.teacheracademy.eu/blog/online-apps-tools-for-teaching/>
4. <https://www.european-agency.org/sites/default/files/itresearch2011findings.pdf>


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Signature of Faculty Member


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HOD